

From MTR/SINIR to the European Buyer

A crosswalk between Brazilian waste-traceability instruments and European supply-chain evidence requests

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Declaration of interest. The author is CEO of Ecobraz, a Brazilian company operating in post-use electronic asset collection and destination, and founder of Villanova ESG. The author has a commercial interest in supplier-evidence services. This crosswalk is vendor-neutral, names no product, and is usable by any operator, including competitors.

Abstract

Brazilian exporters and their European buyers routinely talk past each other about waste and post-use asset evidence. The Brazilian operator holds a legally grounded, systemically issued documentary chain — manifests and destination certificates produced inside federal and state government systems — and often cannot explain what it proves. The European buyer asks for "traceability" in the vocabulary of its own reporting and due-diligence obligations and often does not know that the Brazilian document it has been sent is stronger, in its own domain, than anything it would receive from many other jurisdictions. This report provides a **crosswalk**: for each Brazilian instrument, what it actually evidences, which supplier-evidence category it feeds, which class of European request it can partly answer, and — the column that matters most — **what it does not cover and what must be added**. Two secondary findings are reported honestly: several European instruments frequently cited to Brazilian suppliers, notably the deforestation and carbon-border regimes, **are unlikely to apply to a waste-destination service at all**, and the dominant real gap is not a missing document but a missing **join** between documents that already exist.

Keywords: MTR; SINIR; certificate of final destination; CDF; supplier evidence; EU–Brazil; CSDDD; ESRS E5; GRI 306; waste traceability; crosswalk; due diligence.

1. Purpose and how to read this

This report answers one operational question: **a Brazilian company already holds the documents its own law requires — what part of a European buyer's request do those documents already answer, and what remains open?**

It does not restate the legal text of any instrument; those texts move. It maps at the level of **what evidence an instrument produces**, which is stable, onto the **seven evidence categories** of the Supplier Evidence Maturity Model (DOI 10.5281/zenodo.21445455) and onto the **classes of European request** that recur.

Three cautions before the tables.

- **A crosswalk is not a compliance opinion.** That an instrument *feeds* a category does not mean any specific buyer will accept it, and does not determine whether any obligation applies to a given company. That determination is legal and case-specific.
- **This maps the corporate (B2B) post-use electronic asset flow.** Household reverse-logistics obligations follow a different regime and are noted only where they create a common misunderstanding.
- **Marked ⚠ items** are those where the underlying research base explicitly recorded uncertainty — typically an article number, a phase-in date, or an instrument under revision. They are flagged rather than silently smoothed over.

2. The Brazilian side: what each instrument actually evidences

Brazilian instrument	What it is	What it evidences	What it does not evidence
National Solid Waste Policy (Law 12.305/2010)	The framework statute: environmentally adequate destination, shared responsibility, waste management plans	That a duty of adequate destination exists and to whom it attaches	Performance of that duty in any specific case
MTR — waste transport manifest, in the federal SINIR system (Ordinance MMA 280/2020) ⚠ <i>under revision; verify version in force</i>	Mandatory nationwide since 2021 for generators subject to a management plan; online, self-declared	The generator → transporter → destination movement of a stated waste and mass, on a date, with each party identified by tax number	That the material was treated as intended after receipt
CDF — Certificate of Final Destination, issued in SINIR	Issued exclusively by the licensed destination operator , signed by the responsible technical officer, linked to the underlying MTRs, bearing a unique system identifier	Quantity received and the technology applied to its destination; the closing link of the chain; the probative document of adequate destination	Anything about carbon; anything about material not covered by the linked manifests
Operating Licence (LO) of the destination operator	State environmental licence	That the receiving facility is authorised to operate	That it operated within the licence on a given date
CADRI (São Paulo) and state equivalents	Prior authorisation to send hazardous and certain other classes to a given destination	That the specific movement was pre-authorised	Uniformity: each state has its own system and rules
Brazilian Waste List (IBAMA IN 13/2012) + ABNT NBR 10004 classification ⚠ <i>verify version in force</i>	Waste coding and hazard classification (Classes I / II-A / II-B)	A standardised description and hazard class of the material	Equivalence with European waste codes — the codes do not map one-to-one
ABNT NBR 16156	Standard for reverse-manufacturing activity, including traceability and information security	That a certifiable management standard exists for the dismantling activity	That any given operator holds or applies it
Annual declarations (RAPP/IBAMA; state inventories, e.g. DMR/SIGOR in SP)	Periodic regulatory reporting	Aggregate volumes declared to authorities	Lot-level facts
Voluntary certifications (R2v3, e-Stewards, WEEELABEX/EN 50625 ⚠ <i>verify sub-parts</i>)	Private third-party schemes	Independent assessment of the operator's management system and downstream due diligence, within a stated scope	A legal obligation in Brazil — they are differentiators, not requirements

The structural strength, rarely used commercially. The CDF is issued **inside a government system**, by a **licensed** operator, **linked** to the manifests, with a **unique identifier**. That combination — third-party issuance point, licence anchor, and an identifier that can be checked in the issuing system — is precisely the property that makes a document defensible to someone who does not trust the supplier. Most Brazilian exporters send this document as a PDF attachment and never tell the buyer that it is checkable at source. That is a communication failure, not an evidence failure. (On the general properties that make a destination certificate verifiable, and the limits of verification, see the companion record, DOI 10.5281/zenodo.XXXXXXX.)

The structural weakness. Every one of these instruments is a **waste-destination** instrument. None of them evidences carbon, none evidences data sanitisation, and none is written to be read by someone outside Brazil.

3. The European side: which requests actually arrive

Grouping by the class of request rather than by instrument, because the instruments move and the requests do not.

Class of request	Typical driver	What the buyer is trying to satisfy
Due-diligence	Corporate sustainability due diligence obligations ⚠️ <i>transposition and phase-in have been under revision; verify</i>	Its own duty to identify and address adverse impacts in its chain — needs governance and chain evidence , not a certificate
Circularity / waste reporting	ESRS E5 under the EU corporate sustainability reporting regime ⚠️ <i>scope and timetable under the "Omnibus" revision; verify; GRI 306 for voluntary reporters</i>	Mass by destination method , per period — the single request Brazilian documents answer best
Climate reporting	ESRS E1; IFRS S2 as adopted in the buyer's jurisdiction	Scopes 1/2/3 emissions on a GHG-Protocol basis — an inventory figure, not an avoided-emissions figure
Data protection / asset disposal	The buyer's own security and personal-data obligations	Documented sanitisation of data-bearing media, with attribution
Shipment legality	The EU regime on transboundary waste shipments ⚠️ verify which instrument is currently in force — this regime has been undergoing replacement ; and, for equipment shipped as used rather than as waste, the used-EEE/WEEE distinction under the WEEE Directive (2012/19/EU), which turns on invoice, contract and test documentation	That material crossing a border is lawfully classified and lawfully moved
Claim scrutiny	Directive (EU) 2024/825, applicable from 27 September 2026	That no product is labelled climate/carbon neutral on the basis of offsetting — this constrains what may be said , it is not an evidence request

3.1 Two instruments frequently invoked that probably do not apply

Stating this plainly is more useful to an exporter than adding them to a checklist.

- **The EU deforestation-free products Regulation** ((EU) 2023/1115) covers specified commodities and derived products. A **post-use electronic asset destination service is not within that scope**. An exporter of such services being asked for plot-level geolocation is being asked the wrong question — usually because the buyer applied a generic supplier questionnaire. ⚠️ Verify scope and dates, which have been amended; and note that a company with *other* product lines may be in scope for those.
- **The Carbon Border Adjustment Mechanism** ((EU) 2023/956) applies to importers into the EU of specified goods — cement, iron and steel, aluminium, fertilisers, electricity, hydrogen. A destination service is not an import of those goods. **One genuine edge case deserves attention:** an operator that recovers and *sells metal fractions into the European market* may be touching a covered goods category through that sales channel rather

than through the service. ⚠ This must be verified against the current phase and the actual commercial arrangement; it is raised here as a question to ask, not as a conclusion.

The practical consequence. An exporter that answers "not applicable, and here is why" — in writing, with the reasoning — is displaying better governance than one that fabricates an answer. A buyer's questionnaire is not a statute.

4. The crosswalk

Reading the table: **Feeds** names the SEMM evidence category (DOI 10.5281/zenodo.21445455, §5) the instrument supplies; **Partly answers** names the class of European request from §3; **Gap** is what remains open.

Brazilian instrument	Feeds (SEMM category)	Partly answers	Gap — what must be added
MTR	Identity & traceability; Chain of custody	Circularity/waste reporting; Due-diligence (as chain evidence)	Portuguese-only; Brazilian waste codes; per-movement, not per-period. Needs aggregation and a translated field mapping
CDF	Environmental destination; Chain of custody	Circularity/waste reporting (mass by destination method — the strongest single match); Due-diligence	Does not state the method category in the buyer's vocabulary ; does not carry a mass balance; buyer usually does not know it is checkable at source
Operating Licence + CADRI / state authorisation	Governance & accountability; Environmental destination	Due-diligence	Validity at the date of the movement must be evidenced, not merely the licence's existence
Waste List code + NBR 10004 class	Identity & traceability	Circularity/waste reporting	No one-to-one mapping to European waste codes ; needs an explicit, documented correspondence with its own limits stated
NBR 16156 / voluntary certification (R2v3, e-Stewards, WEEELABEX)	Governance & accountability; Data security	Due-diligence; Data protection	Scope of the certificate must be checked against the specific operation ; a certificate held by the group does not cover every site
Data-destruction record (contractual/normative — not a legal instrument in Brazil)	Data security	Data protection	Must be attributable, dated, per-lot or per-device, and issued by the party that performed it
Annual declarations (RAPP / state inventory)	Governance & accountability	Circularity/waste reporting	Aggregate only; cannot answer a lot-level question
— no Brazilian instrument —	Carbon / CO₂e	Climate reporting	Nothing in the Brazilian waste framework evidences carbon. It requires a separate quantification with an explicit baseline, and it produces <i>avoided emissions</i> , which are not an inventory figure — see the companion record on the declaration protocol
— no Brazilian instrument —	Origin & geolocation	Deforestation-type requests	Not applicable to this flow (§3.1)

5. The six gaps, in order of how often they decide an assessment

1. **The join, not the document.** The chain exists across MTR, CDF, licence and invoice, and is usually filed by client and month rather than by lot and event. A buyer asking "what happened to *this* delivery" cannot be answered from a complete file that is organised the wrong way. This is the single most common failure and the cheapest to fix.
2. **Language and code.** Documents are in Portuguese, using Brazilian waste codes and Brazilian legal categories. A buyer-readable file needs a translated field mapping — **with the non-equivalences stated**, since the coding systems do not correspond one-to-one.
3. **Verifiability not communicated.** The strongest property of the Brazilian file — issuance inside a government system, under a licence, with a checkable identifier — is almost never explained to the buyer.
4. **Mass balance absent.** Documents state received quantities; they rarely reconcile inbound against outbound by destination. Without that, a destination percentage cannot be substantiated whatever the certificates say.
5. **Carbon confused with destination.** A destination certificate is repeatedly presented as though it evidenced a carbon outcome. It does not, and the confusion is now a legal exposure as well as an evidential one.
6. **Licence validity as of the date.** Holding a licence today is not evidence that it was valid on the date of the movement being evidenced.

None of these six requires a new legal instrument. All six are fixed in how existing evidence is structured, joined and explained.

6. What a buyer-readable file contains

Derived from §4 and §5; this is the second citable artifact of the report.

Element	Source	Note
Period, client entity, and statement of coverage	Operator's records	Must state what is <i>not</i> covered: material disposed of through other channels; manifests still open at the cut-off
Mass by destination method, per period	CDF	The element that maps most directly to European circularity reporting
Lot-level chain, on request	MTR ↔ CDF ↔ invoice, joined by lot	The answer to "what happened to this delivery"
Licence and authorisation references, valid at the movement date	LO, CADRI/state authorisation	Include the reference, not a scan alone
Waste code correspondence table	Brazilian Waste List / NBR 10004, mapped	State the limits of the mapping
Verification note	SINIR identifier and how to check it	Turns a PDF into something the buyer can confirm without the supplier
Data-sanitisation records	Operator/subcontractor	Attributable and dated
Carbon — if any	Separate document, separate method	Labelled as avoided emissions, never merged into the destination file
Exceptions	Operator's records	Non-conformities, open manifests, unreconciled residual

An operator that can produce this file for an arbitrary period, without opening a special project, is at Level 3 of the maturity model for the categories concerned.

7. Limitations

This is a version 1.0 conceptual mapping, not peer reviewed, not empirically validated, and **not legal advice**. It maps instruments to evidence classes; it does not determine whether any obligation applies to any company. Brazilian state rules differ and only São Paulo's arrangement is referenced by name as an example. Several instruments on both sides are under revision, replacement or phased entry, and every ⚠ item must be verified against current official sources before use. Article and sub-paragraph numbers are deliberately omitted where the underlying research base recorded uncertainty about them. The report addresses the **provability** of destination, not environmental performance, and produces no carbon figure or claim.

8. How to cite

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(DOI to be inserted after deposit.)

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Instruments are cited for orientation. Verify exact titles, numbers, dates and status at the official source before any operational decision.

1. Brazil. Law 12.305/2010 — National Solid Waste Policy, and its regulation (Decree 10.936/2022).
2. Brazil. Ordinance MMA 280/2020 — national MTR and the SINIR system. ⚠️ Verify the version in force.
3. Brazil. Decree 10.240/2020 — reverse logistics for household electrical and electronic products (corporate flows excluded). ⚠️ Verify.
4. Brazil. IBAMA Normative Instruction 13/2012 — Brazilian Waste List; ABNT NBR 10004 — waste classification. ⚠️ Verify version in force.
5. Brazil. ABNT NBR 16156 — reverse manufacturing of electrical and electronic waste.
6. São Paulo State. CETESB/SIGOR; CADRI and operating licence regime. ⚠️ State rules differ; verify per state.
7. Directive 2012/19/EU (WEEE), including the used-EEE/WEEE distinction. ⚠️ Verify.
8. EU regime on transboundary shipments of waste. ⚠️ **Verify which instrument is currently in force.**
9. Directive (EU) 2024/825 — empowering consumers for the green transition; applicable from 27 September 2026. ⚠️ Verify transposition.
10. Directive (EU) 2024/1760 — corporate sustainability due diligence. ⚠️ Verify transposition and phase-in.
11. Regulation (EU) 2023/1115 (deforestation-free products) and Regulation (EU) 2023/956 (CBAM) — cited in §3.1 as instruments unlikely to apply to this flow. ⚠️ Verify scope and dates.
12. ESRS E1 and E5; GRI 306: Waste (2020); IFRS S1/S2. ⚠️ Verify scope and timetable, under revision.
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